Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: A Phase 1/2A Study Evaluating the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Anti-Tumor Activity of PF-07220060 As a Single Agent and As Part of Combination Therapy in Participants With Advanced Solid Tumors **Short Title/Acronym:** Atirmociclib (PF-07220060) Phase 1/2A in Advanced Solid Tumors **Protocol Number:** C4391001 **NCT Number:** NCT04557449 **Sponsor Name:** Pfizer **Investigational Product:** Atirmociclib (PF-07220060) - CDK4-selective inhibitor **Phase of Development:** Phase 2 (Phase 1/2A) **Trial Status:** Active, not recruiting **Medical Monitor:** Pfizer Clinical Operations Lead

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and maximum tolerated dose (MTD) of atirmociclib (PF-07220060) administered as a single agent and in combination with endocrine therapy or enzalutamide, and to select the recommended dose for expansion. **Secondary Objectives:** To evaluate the preliminary anti-tumor activity (including Objective Response Rate [ORR], Clinical Benefit Rate [CBR], Progression-Free Survival [PFS], and Duration of Response [DOR]), pharmacokinetics (PK), and pharmacodynamics (PD) of the study drug.

2.2 Background & Rationale To address the need for targeted therapies in advanced solid tumors, specifically hormone receptor (HR)-positive breast neoplasms, prostate cancer, adenocarcinoma of the lung (NSCLC), liposarcoma, and colorectal cancer. Atirmociclib (PF-07220060) is a novel, potent, oral CDK4-selective inhibitor designed with significant sparing of CDK6. This mechanism aims to maintain high anti-cancer efficacy while minimizing CDK6-mediated toxicities like severe neutropenia, which frequently limit dose intensity in first-generation CDK4/6 inhibitors.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose Escalation and Dose Expansion cohorts) **Blinding:** Open-label **Randomization:** Nonrandomized

3.2 Planned Enrollment & Duration Target N: 362 participants across multiple cohorts. **Study Structure:** The study consists of Part 1 (Dose Escalation: 1A-1F), Part 2 (Dose Expansion: 2A-2E), and dedicated China and Japan monotherapy cohorts. **Number of**

Sites: Multicenter (United States and 7 other countries)
Estimated Study Start: 22-Sep-2020 **Estimated Primary Completion:** Active, not recruiting

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years, all genders, with an Eastern Cooperative Oncology Group (ECOG) Performance Status 0 or 1, and adequate renal, liver, and bone marrow function. 2. Confirmed diagnosis of advanced solid tumors. Specifically targeting: * Breast Neoplasms (HR+/HER2- and HR+/HER2+). * Prostate Cancer, including metastatic castration-resistant prostate cancer (mCRPC). * Adenocarcinoma of Lung (NSCLC), Colorectal Cancer (CRC), Liposarcoma, or tumors with previously confirmed CDK4 or CCND1 amplification. 3. Participants must be refractory to or intolerant of existing therapies known to provide clinical benefit. 4. Prior treatment specifics depend on the cohort: * **Part 1:** Progressed on ≥ 1 line of standard of care (SOC) including endocrine therapy and/or CDK4/6 inhibitors for HR+/HER2- BC; ≥ 1 prior HER2 targeted therapy for HR+/HER2+ BC; or ≥ 2 lines of SOC for other tumors. * **Part 2B:** Treatment-naive for advanced disease. * **Part 2C:** Progressed on or within 12 months of adjuvant AI or tamoxifen, or within 1 month of prior AI. * **Part 2D:** Prior abiraterone; must be enzalutamide and CDK4 inhibitor naive (0-1 line of chemotherapy allowed).

4.2 Exclusion Criteria 1. Known active, uncontrolled, or symptomatic Central Nervous System (CNS) metastases, carcinomatous meningitis, or leptomeningeal disease. 2. For Part 1D: Participants who have had a gastrectomy or have dietary restrictions that preclude a 10-hour overnight fast or consumption of a high-fat, high-calorie meal. 3. For Part 2C: Prior treatment with any CDK inhibitor, fulvestrant, everolimus, or any agent whose mechanism of action is to inhibit the PI3K-mTOR pathway. 4. Other active malignancy within 3 years prior to entry, except adequately treated basal cell or squamous cell skin cancer, or carcinoma in situ. 5. Major surgery or radiation within 4 weeks; last anti-cancer treatment within 2 weeks; or participation in other studies involving investigational drugs within 4 weeks prior to study intervention. 6. Pregnant or breastfeeding female participants. 7. Active inflammatory gastrointestinal (GI) disease, known diverticular disease, or previous gastric resection/lap band surgery.

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atirmociclib (PF-07220060) at escalating doses administered as a single agent (Part 1A, China/Japan cohorts) or in combination. **Combination Therapies:** Administered with letrozole (Parts 1B, 2B), fulvestrant (Parts 1C, 2A, 2C, 2E), or enzalutamide (Parts 1F, 2D). Part 1D evaluates food effects, and Part 1E evaluates the drug-drug interaction (DDI) effect of PF-07220060 on the pharmacokinetics of midazolam. **Route of Administration:** Oral **Duration of Treatment:** Administered in 28-day cycles until disease progression, unacceptable toxicity, or study withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Endocrine therapies including Letrozole or Fulvestrant, or Enzalutamide, as defined by specific study arms. Midazolam administration is protocol-specified in Part 1E. **Prohibited:** Concurrent investigational agents or other anti-cancer therapies outside the specified combinations.

---

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Days -28 to 0 | Informed Consent, Medical History, Baseline Tumor Assessment | | **Cycle 1** | Days 1, 15 | First Dose, Intensive PK/PD sampling, Safety monitoring | | **Subsequent Cycles** | Day 1 of each 28-day cycle | Safety Labs, Efficacy Assessment (Tumor Imaging), PK checks | | **End of Study** | Up to ~24 months | Final Vital Signs, Exit Interview, Follow-up for disease progression/survival |

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. **Dose-Limiting Toxicities (DLTs):** Number of participants with DLTs in the Dose Escalation Portion (First cycle [28 days] for Parts 1A, 1B, 1C, 1F) to establish the Maximum Tolerated Dose (MTD). 2. **Adverse Events:** Incidence of clinically significant treatment-emergent adverse events (TEAEs). 3. **Safety Labs:** Incidence of clinically significant laboratory abnormalities.

7.2 Secondary & Exploratory Endpoints 1. **Pharmacokinetics:** Single Dose Maximal concentration ($Cmax$), Time to Maximum Plasma Concentration ($Tmax$), and Area Under the Plasma Concentration Versus Time Curve from Time Zero to the Last Sampling Time Point Within the Dose Interval ($AUClast$) in the Dose Escalation and Dose Finding portion. 2. **Preliminary Anti-Tumor Efficacy:** Objective Response Rate (ORR), Clinical Benefit Rate (CBR), Progression-Free Survival (PFS), and Duration of Response (DOR).

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for TEAEs and clinically significant safety laboratory abnormalities. **Serious Adverse Events (SAEs):** Reported promptly per standard regulatory safety protocols (typically within 24 hours of site awareness). **Data Monitoring Committee (DMC):** Internal/External safety review committee to evaluate dose-escalation cohorts and clear dose progression.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose of study drug) and Efficacy Evaluable Set (patients with measurable disease at baseline and post-baseline scans). **Sample Size Justification:** Total targeted enrollment of 362 participants. Adaptive design typical for Phase 1/2A studies; driven by the classical 3+3 or modified continual reassessment method for dose escalation, and specific clinical thresholds for expansion cohorts. **Handling of Missing Data:** Standard descriptive statistics; missing data for efficacy endpoints typically handled via censoring at the last evaluable assessment.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical site monitoring visits to verify adherence to protocol, source data verification, and drug accountability. **Audit Trail:** Electronic Case Report Form (eCRF) utilizing secure, FDA CFR 21 Part 11 compliant data capture systems.

---

11. Study Identifiers & Governance

Primary ID: C4391001 **Registry Number:** NCT04557449 **Sponsor/Lead Organization**: Pfizer **Study Title:** Study to Test the Safety and Tolerability of PF-07220060 in Participants With Advance Solid Tumors **Official Regulatory Title:** A Phase 1/2A Study Evaluating the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Anti-Tumor Activity of PF-07220060 As a Single Agent and As Part of Combination Therapy in Participants With Advanced Solid Tumors

---

12. Clinical Framework (Oncology Specific)

Primary Condition: Advanced Solid Tumors (Breast Neoplasms, Prostate Cancer, Liposarcoma, and Adenocarcinoma of Lung) **Diagnosis Threshold:** Histologically confirmed advanced, recurrent, or metastatic disease **Medical Methodology:** Targeted CDK4 inhibition as single agent or combined with endocrine therapy/enzalutamide **Optimization Target:** Maximal tumor response with minimized CDK6-mediated hematologic toxicity

---

13. Study Procedures & Methodology

Management Pathway: Stepwise dose escalation followed by dose expansion in specific tumor types. **Education Protocol (HCP):** Investigator protocol training, safety management (especially regarding hematologic and GI toxicities). **Education Protocol (Patient):** Informed consent, dosing diaries, and fasting/dietary instructions for specific cohorts (e.g., Part 1D food effect). **Quality Audit Mechanism:** Independent central review of imaging (where applicable) and regular sponsor auditing.

---

14. Observation & Follow-Up Schedule

Total Duration: Follow-up for progression and survival (up to ~24 months or until progression) **Onsite Visit Cadence:** Every 2-4 weeks (Days 1 and 15 of Cycle 1, then Day 1 of subsequent 28-day cycles) **Remote Monitoring Frequency:** As needed based on site SOPs and patient safety profiles **Checklist Review Interval:** End of each 28-day treatment cycle

---

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]